Pyruvate Oxidation and the Krebs Cycle

Unit 3B Slides 17 through 26 Honors High School Biology

Slide 17: Steps 2 and 3

Glycolysis ends with two pyruvate molecules. If oxygen is available, cellular respiration continues with pyruvate oxidation and the Krebs cycle. These stages occur in the mitochondrial matrix and prepare high-energy electrons for the electron transport chain.

Although only a small amount of ATP is produced directly, these stages are essential because they transfer much of pyruvate's remaining energy to NADH and FADH2. Those carriers will deliver electrons to the ETC, where most ATP is produced.

Slide 18: Step 2: Oxidation of Pyruvate

Pyruvate still contains substantial chemical energy. In eukaryotic cells, each pyruvate enters the mitochondrial matrix during aerobic respiration. Because glycolysis produces two pyruvate per glucose, every reaction shown in this stage occurs twice.

This step is often called the link reaction because it connects glycolysis to the Krebs cycle. It converts three-carbon pyruvate into two-carbon acetyl-CoA, which can enter the cycle.

Slide 19: Pyruvate Oxidation Steps

First, one carbon is removed from each pyruvate and released as carbon dioxide. This is decarboxylation. Second, electrons and hydrogen are transferred to NAD+, reducing it to NADH. Finally, coenzyme A attaches to the remaining two-carbon acetyl group, producing acetyl-CoA.

Per glucose, pyruvate oxidation produces two acetyl-CoA, two NADH, and two carbon dioxide molecules. Acetyl-CoA enters the Krebs cycle, while NADH stores electron energy for the ETC. No ATP is produced directly during this stage.

Slide 20: Pyruvate Oxidation Video

As you watch, follow the three-carbon pyruvate molecule. Identify when carbon dioxide is removed, when NAD+ becomes NADH, and when coenzyme A attaches. Do not focus on memorizing every enzyme; focus on how carbon and energy are transferred.

Slide 21: Pyruvate Oxidation in Detail

Use the diagram to account for all three carbons in pyruvate. One carbon leaves as carbon dioxide, and the remaining two form the acetyl group carried by coenzyme A. NADH leaves with high-energy electrons and hydrogen.

Carbon dioxide diffuses out of the mitochondrion and the cell. In animals, it enters the bloodstream, travels to the lungs, and is exhaled. This is part of the carbon dioxide measured when respiration rate increases.

Slide 22: Step 3: The Krebs Cycle

The Krebs cycle is also called the citric acid cycle. Hans Krebs helped identify the pathway in 1937. Because one glucose produces two acetyl-CoA molecules, the cycle turns twice for each glucose.

The main goal is not direct ATP production. Acetyl-CoA is repeatedly oxidized, and its electrons reduce NAD+ and FAD to NADH and FADH2. These carriers store energy that the electron transport chain uses. Carbon atoms are removed as carbon dioxide.

Slide 23: Stages of the Krebs Cycle

First, the two-carbon acetyl group joins four-carbon oxaloacetate, forming six-carbon citrate, or citric acid. Coenzyme A is released and can be reused.

Next, citrate is rearranged and oxidized through several reactions. Electrons reduce NAD+ and FAD, carbon dioxide is released, and one ATP is generated per turn through substrate-level phosphorylation. The remaining molecules are then rearranged to regenerate oxaloacetate. Regeneration is essential because oxaloacetate must accept the next acetyl group.

Slide 24: Krebs Cycle Video

While watching, track four ideas: acetyl-CoA joining oxaloacetate, carbon leaving as carbon dioxide, NADH and FADH2 being produced, and oxaloacetate being regenerated. The names of the intermediates matter less than the overall movement of carbon and energy.

Slide 25: Krebs Cycle in Detail

This diagram represents both turns of the cycle per glucose. Each turn begins when acetyl-CoA contributes two carbons. Those carbons are ultimately released as carbon dioxide, while oxaloacetate is regenerated rather than consumed.

Most of the captured energy leaves in NADH and FADH2, not ATP. This explains why the cycle is central to aerobic respiration even though it makes only one ATP per turn. The carriers connect the cycle to the ETC, where their electrons power the production of much more ATP.

Slide 26: Krebs Cycle Summary

For each glucose, the Krebs cycle turns twice and uses two acetyl-CoA. Its energy products are 2 ATP, 6 NADH, and 2 FADH2. It also releases four carbon dioxide molecules. Including the two carbon dioxide molecules from pyruvate oxidation, all six original glucose carbons are released as carbon dioxide.

The NADH and FADH2 move to the electron transport chain and donate their high-energy electrons. By the end of glycolysis, pyruvate oxidation, and the Krebs cycle, the cell produces only a small amount of ATP directly, but it accumulates the electron carriers needed for the largest ATP-producing stage.